

Development of *Deep Neural Network* Models for Scoliosis Detection

DualPathNet — a multi-task architecture for automated Cobb angle estimation with uncertainty-aware outputs.

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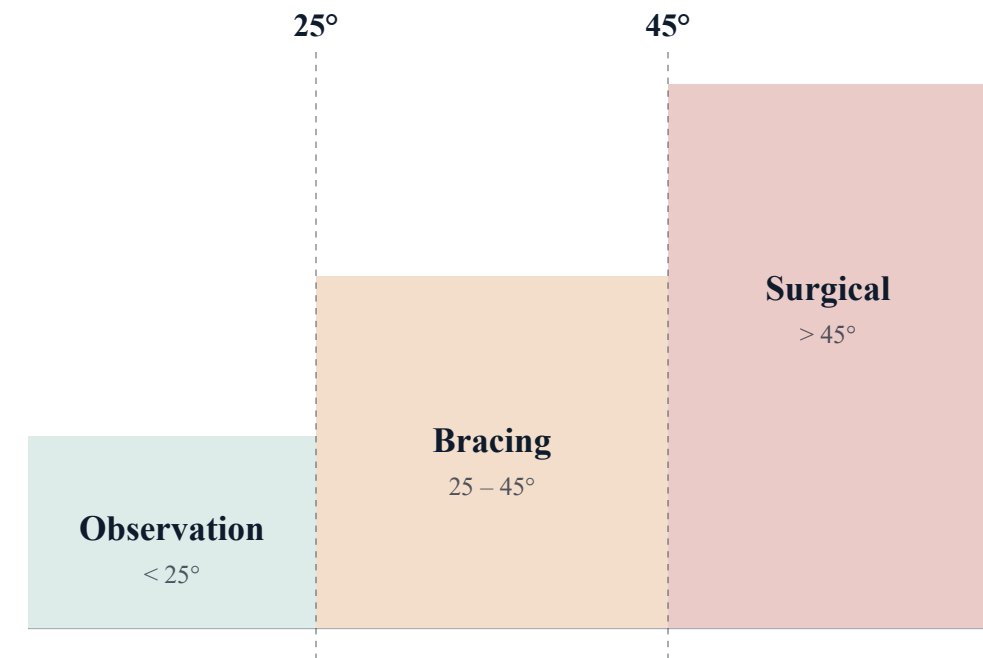
Beibit Abdikenov, PhD

Why automated Cobb angle measurement matters

CLINICAL CONTEXT

- ▶ Scoliosis is a lateral spinal curvature exceeding **10°** on AP radiograph; affects **2–3%** of adolescents.
- ▶ The **Cobb angle** drives every treatment decision — observation, bracing, or surgery.
- ▶ Manual measurement is slow and reader-dependent: inter-observer error **3–5°**, intra-observer $\approx 3^\circ$.
- ▶ A **5°** error can shift a patient across treatment thresholds — hence the need for reliable automation.

TREATMENT THRESHOLDS



A 5° error near a threshold is the difference between observation and intervention.

Purpose and logic of the study

To develop and experimentally evaluate a *multi-task neural network* for automated Cobb angle estimation from spinal radiographs, incorporating *dual-path angle prediction* and *uncertainty-aware outputs*.

WHAT IS PREDICTED

Three regional Cobb angles

PT, MT, and TL/L outputs are treated as anatomical curve slots, not only sorted angle magnitudes.

HOW IT IS PREDICTED

Direct + geometric paths

A global regression path is fused with local vertebra-tilt geometry through a learned gate.

WHY IT MATTERS

Decision-support reliability

Per-angle uncertainty is evaluated as a signal for deferring difficult cases to expert review.

Three testable hypotheses

PRIMARY H1

Accuracy below 2.5°

A dual-path multi-task neural network can achieve mean absolute error below **2.5°** on leakage-mitigated SpinalAI2024 validation and holdout splits under the PT/MT/TL-L protocol.

SECONDARY H2

Uncertainty ranks risk

Learned per-angle uncertainty can rank high-error cases well enough to support **selective expert review** instead of treating every prediction as equally reliable.

EXPLORATORY H3

Clinical-domain adaptation

A model pre-trained on synthetic SpinalAI2024 images can be adapted to real AASCE19 radiographs through limited fine-tuning, while zero-shot transfer remains constrained by domain shift.

Scope of the investigation

OBJECT OF RESEARCH

Automated radiographic assessment of spinal curvature

The clinical task of measuring Cobb angles from **digital anteroposterior spinal radiographs** — the input domain of the study, including image quality variability, vertebral morphology, and curve topology.

PT · MT · TL/L — three regional curve slots

SUBJECT OF RESEARCH

A multi-task convolutional neural network for joint landmark detection & angle estimation

Specifically, the **DualPathNet** architecture — including its global / local pathways, learned fusion gate, varying input resolutions (512 & 1024 px), and heteroscedastic uncertainty head.

What prior work establishes — and what it leaves open

LITERATURE BLOCK	USED SOURCES	WHAT THEY SUPPORT	OPEN GAP	THESIS RESPONSE
Clinical measurement	Cobb; Morrissy; Reamy & Slakey; Weinstein	Cobb angle thresholds and reader variability define clinical risk.	Errors near 25°/45° can change treatment class.	Threshold and uncertainty analysis.
Automated Cobb benchmarks	AASCE2019; Seg4Reg+; VLTENet; CurvNet	Landmarks, segmentation and direct regression are all viable.	Protocols differ; regional slot semantics are often weak.	Strict PT/MT/TL-L three-curve output.
Multi-task learning	Zhang & Yang; Crawshaw; clinical multi-task models	Intermediate anatomical tasks can regularise final prediction.	Task competition can destabilise training.	Composite loss and staged curriculum.
Uncertainty & reporting	Kendall & Gal; Lakshminarayanan; CLAIM	Medical AI needs reliability estimates, not only point MAE.	Calibration and risk ranking are rarely analysed for Cobb angle.	Heteroscedastic head + risk-coverage.
Synthetic-to-clinical data	SpinalAI2024; AASCE19; leakage-control literature	Synthetic data can scale annotation, but domain shift must be tested.	Near-duplicate leakage and target-domain transfer remain risks.	Proxy grouping, filtering and AASCE fine-tuning.

What is new in the developed model and evaluation

01 METHOD

Proposed slot-aware dual-path prediction

Direct PT/MT/TL-L regression is fused with a geometry-derived signal from vertebral tilts. The direct path preserves anatomical slots; the geometric path regularises the angle estimate.

03 EVALUATION

Experimentally confirmed resolution paradox

The experiments compare landmark error and Cobb angle MAE together, showing that better high-resolution keypoints do not automatically improve regional angle accuracy.

02 LOSS DESIGN

Introduced scale-normalised coordinate loss

Normalised coordinates are multiplied by a fixed 512 px reference scale before smooth L1, keeping coordinate-loss gradients comparable across 512 px and 1024 px inputs.

04 RELIABILITY

Integrated uncertainty for triage

A per-angle variance head is trained with the fused output and evaluated as a risk-ranking signal: the lowest-uncertainty 10% reaches 1.77° MAE.

These novelty points directly answer the gaps from the literature slide: regional semantics, scale fairness, resolution effects and prediction reliability.

Dataset structure, features and split protocol

DATA SOURCE	RAW STRUCTURE	FEATURES / LABELS USED	ROLE IN THIS THESIS
SpinalAI2024 primary synthetic	AP radiograph-like PNG images, COCO-style vertebral quadrilaterals, per-image angle records and metadata.	Image tensor; vertebral centres; corner offsets; per-vertebra tilts; vertebra count; regional PT / MT / TL-L angles.	Model development, validation, holdout testing, ablations, uncertainty and threshold analysis.
AASCE19 clinical external	609 real AP spinal radiographs with vertebral corner annotations from the AASCE challenge.	Clinical image tensor; landmark-derived geometry; Cobb angle targets under the challenge protocol.	Domain adaptation evidence: zero-shot transfer, preprocessing and fine-tuning experiments.

LEAKAGE-MITIGATED SPINALAI2024 SPLIT

- 01

Proxy grouping: 139 groups from filename stems because clinical patient IDs are unavailable.
- 02

Similarity filter: perceptual + cosine criteria removed 323 near-duplicates; 19,677 records retained.
- 03

Group-disjoint split: no proxy group shared across train, validation and test partitions.

EXPORTED SPLIT COUNTS

TRAIN

15,742

VALIDATION

1,956

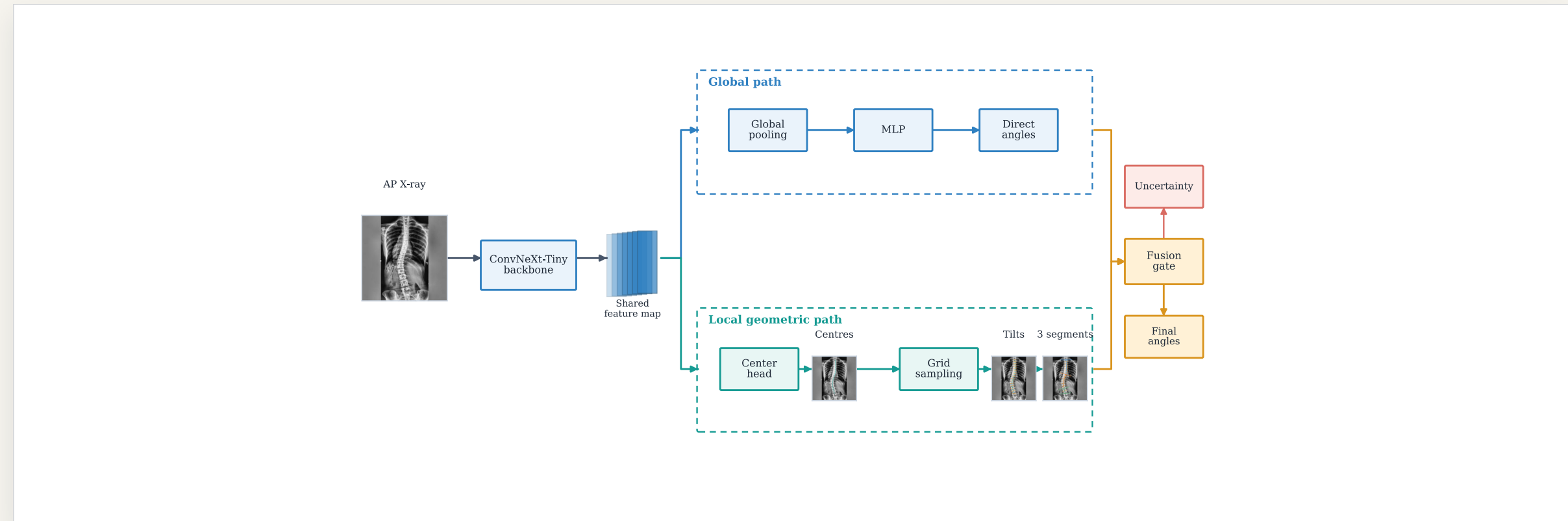
TEST

1,978

Final holdout metrics use 1,977 complete labelled test cases.

SOLUTION MODEL ARCHITECTURE

DualPathNet architecture: global semantics + local geometry



GLOBAL PATH

Learns image-level scoliosis context and directly regresses regional Cobb-angle slots.

LOCAL PATH

Uses vertebral centres, corners and tilts to preserve geometric interpretability.

FUSION

Combines direct and geometric estimates; uncertainty flags cases for expert review.

MAIN RESULTS OF THE RESEARCH

Headline numbers — primary hypothesis confirmed

HOLDOUT TEST + TTA

2.482°

± 0.056 | 1,977 evaluated

WITHIN-5° (CURVE)

89.4%

case-level 80.4%

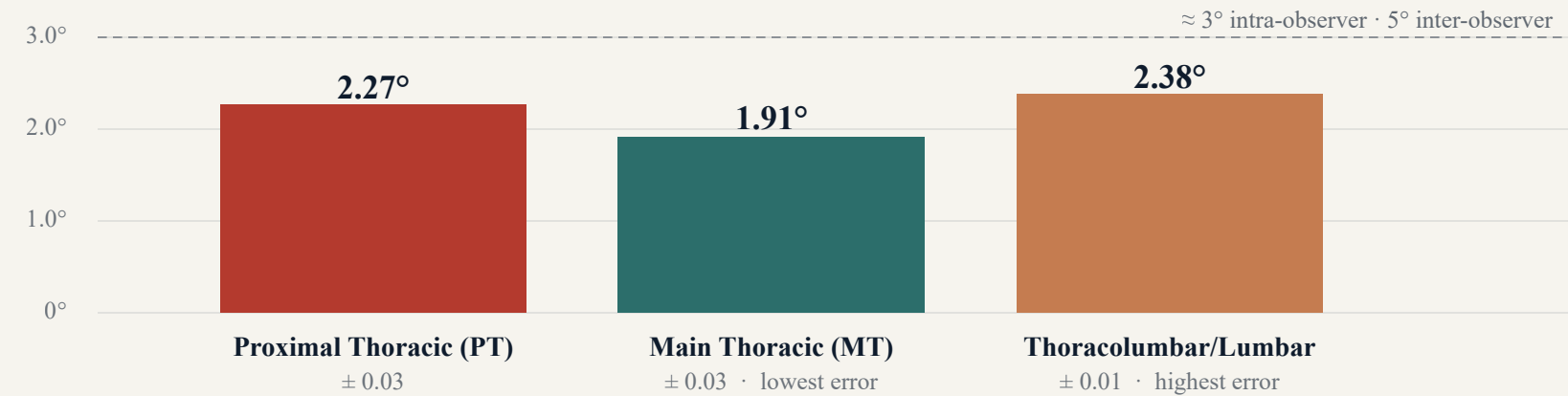
AASCE19 (FINE-TUNED)

2.19°

92.2 % within 5°

PER-CURVE MAE — VALIDATION (512 px, 3 SEEDS)

Main thoracic curve is most accurate



Two X-ray images of a human spine. The left image shows a thoracic curve with a blue line representing the model's output and an orange line representing the ground truth. The right image shows a lumbar curve with a blue line representing the model's output and an orange line representing the ground truth. Both images show the model's output as a series of connected line segments, indicating regional Cobb construction.

FIGURE: MODEL OUTPUT

Numerical accuracy is paired with inspectable anatomy.

The reported MAE is not a black-box score only: the output exposes regional Cobb construction for PT, MT and TL/L confirmation.

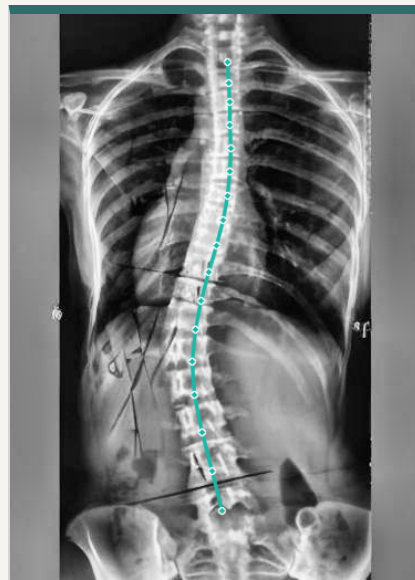
PREDICTION EXPLAINED VISUALLY

From an X-ray to interpretable Cobb-angle predictions



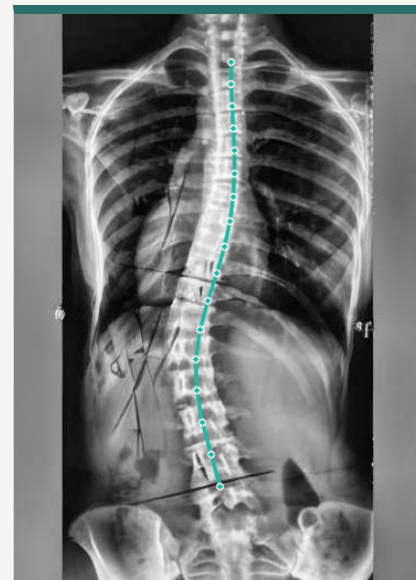
01 INPUT

Original AP spinal radiograph after preprocessing.



02 CENTRES

Model localises vertebral centre candidates.



03 FILTERED

Geometry is corrected into an ordered spine axis.



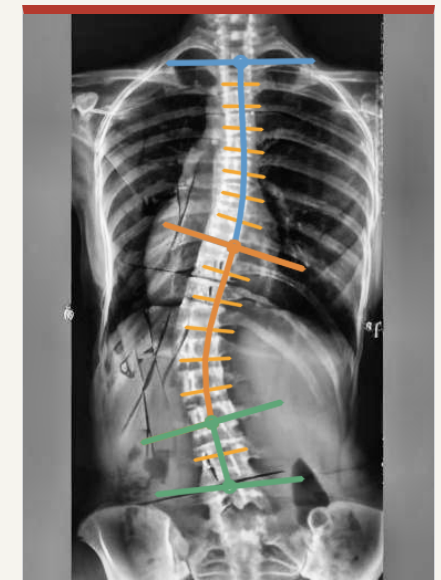
04 TILTS

Local head estimates each vertebra orientation.



05 PRIMARY

Primary Cobb angle is derived from the extremal tilted vertebrae.



06 REGIONS

PT, MT and TL/L outputs remain clinically inspectable.

This visual chain gives the committee the key evidence: the network is not only returning a number, it exposes anatomical landmarks, vertebral tilts and region-specific Cobb-angle construction for expert review.

SCIENTIFIC RESULT

What helps — and what doesn't

COMPONENT EFFECT ON MAE

Dual-path fusion contributes the largest single drop

CHANGE

 Δ MAE (°)

Dual-path fusion

global + geometric paths combined via learned gate

-0.11

Tilt auxiliary loss

direct supervision on per-vertebra tilts

-0.10

Extended training

274 epochs vs 180 — diminishing return

-0.06

Resolution 512 → 1024 px

keypoint pixel error halves, but angle MAE degrades

+0.13

0

BACKBONE STUDY (512 px)

ConvNeXt-Tiny wins despite mid-tier capacity

ConvNeXt-Tiny

~28 M params

2.17°

EfficientNet-B0

~5 M params

2.68°

ResNet-34

~21 M params

2.85°

ConvNeXt's 7×7 depth-wise convolutions capture spinal-curve span better than 3×3 ResNet blocks at the same resolution.

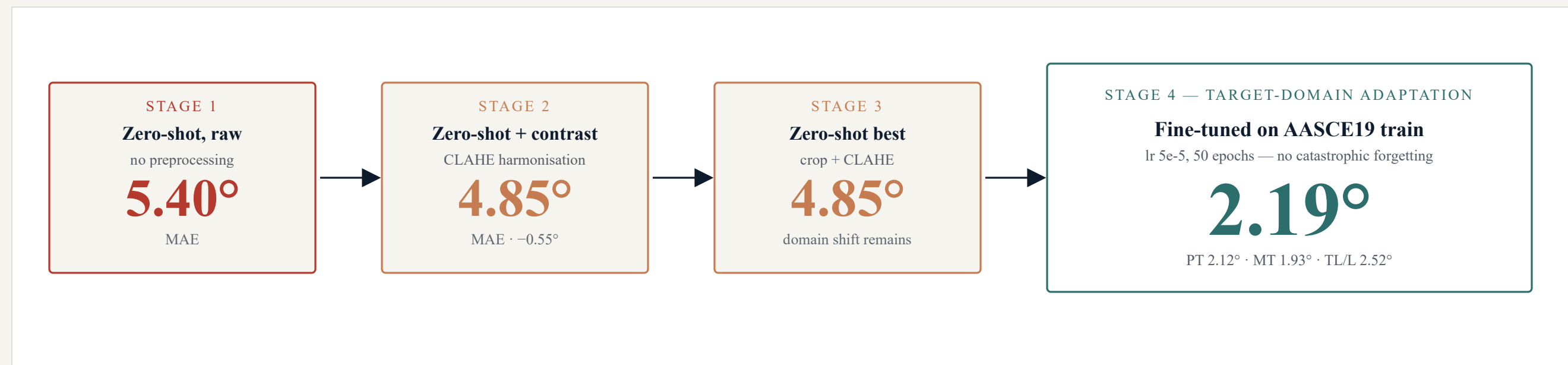
PARAMETER EFFICIENCY > RAW CAPACITY

Resolution paradox: doubling resolution halves keypoint error but hurts angle MAE — the “more pixels = better” assumption fails for global pooling.

DOMAIN ADAPTATION EVIDENCE

Synthetic-trained → clinical AASCE19

SpinalAI2024 is fully synthetic. To assess clinical adaptability, the model was tested on the independent **AASCE19 challenge** (480 labelled real radiographs) under zero-shot and fine-tuned settings.



DOMAIN SHIFT IS REAL

Synthetic mathematical perfection ≠ clinical noise / acquisition artifacts / pelvic structures. Zero-shot MAE ≥ 4.85°.

CONTRAST > CROP / TTA

Most of the zero-shot gain (5.40 → 4.85°) came from CLAHE alone. TTA & cropping had marginal effect.

FINE-TUNING WORKS

92.17% of predicted angles within 5° of expert.
83.16% of patients have all three curves within 5°.

LIMITATIONS

What this thesis does *not* claim

L1

Synthetic-only primary dataset

All 20,000 SpinalAI2024 records flagged `is_synthetic = 1`. Internal holdout performance cannot establish clinical-only generalisation. Only AASCE19 transfer touches real radiographs.

DATASET

L2

No formal clinical reader study

Model MAE measures deviation from a single reference annotation; **inter-observer variability** measures disagreement between two raters. The two are not equivalent without a prospective reader-study comparison.

VALIDITY

L3

Uncertainty under-calibrated at low nominal coverage

Post-hoc λ -scaling restores high-target coverage (95% $\rightarrow$ 92.5%) but leaves nominal intervals $< 50\%$ effectively uncovered — a geometric limit at our MAE level. Clinical use of intervals would require conformal or non-parametric calibration.

CALIBRATION

Published article evidence before conclusion

PUBLICATION RECORD

A dual-path multi-task framework for strict three-curve Cobb angle estimation in idiopathic scoliosis

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JOURNALScientific Journal of Astana IT University

VOLUME25, pp. 234–242

PUBLISHED30.03.2026

DOI10.37943/25XRRS8272

PUBLISHED EVIDENCE

Scientific Journal of Astana IT University
ISSN (P): 2707-9031 ISSN (E): 2707-904X
VOLUME 25, MARCH 2026

DOI: 10.37943/25XRRS8272

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A DUAL-PATH MULTI-TASK FRAMEWORK FOR STRICT THREE-CURVE COBB ANGLE ESTIMATION IN IDIOPATHIC SCOLIOSIS

Abstract: Adolescent idiopathic scoliosis management depends on reproducible Cobb angle measurement across three clinically defined spinal regions: proximal thoracic, main thoracic, and thoracolumbar/lumbar. Although manual measurement remains the reference standard, it is observer-dependent and time-consuming, with inter-observer variability exceeding five degrees even among experienced readers. Most automated deep learning approaches target a single dominant curve or use unconstrained outputs, which limits their applicability to structured clinical workflows requiring strict regional assignment. This study presents a dual-path multi-task framework for simultaneous estimation of all three regional Cobb angles from posteroanterior spinal radiographs. The architecture integrates a ConvNeXt-Tiny encoder, vertebral localization heads, direct global angle regression via soft-argmax, and a geometric tilt-aggregation pathway. A learned per-region sigmoid gate fuses the global and geometric pathways, providing a fixed but optimized balance between statistical and anatomical estimation. The model was developed on 21,294 radiographs with leakage-controlled partitioning into training (N = 17,262), validation (N = 2,016), and test (N = 2,016) subsets. Training employed a two-stage curriculum with severity-aware sampling and hard replay for difficult cases. Three independent runs (seeds 42, 52, 62) were ensembled with test-time augmentation. On the primary held-out set (N = 2,015), the ensemble achieved a mean absolute error of 2.24 degrees (proximal thoracic 2.21, main thoracic 1.97, thoracolumbar/lumbar 2.54), with near-zero Bland-Altman bias (0.03 degrees), good-to-excellent intraclass correlation coefficients (0.884–0.971), and 90.4% of predictions within 5 degrees. At the 40-degree treatment threshold, sensitivity was 0.934 and specificity was 0.994. These findings support the feasibility of strict three-curve automation for reader-in-the-loop clinical workflows.

Keywords: Cobb angle estimation; scoliosis; deep learning; multi-task learning; dual-path fusion; medical imaging; curriculum learning; clinical decision support.

Introduction
Adolescent idiopathic scoliosis is the most common spinal deformity in the pediatric population, affecting approximately two to three percent of adolescents worldwide [1, 2]. Clinical management decisions, including observation, bracing, and surgical referral, rely heavily on accurate and reproducible Cobb angle measurement. A five-degree measurement change at critical thresholds can alter the treatment pathway, making measurement consistency a fundamental concern for clinicians [1]. Despite its clinical centrality, the Cobb method remains a manual process susceptible to non-trivial inter- and intra-observer variability, which several reliability studies have documented across different clinical settings and reader populations [1, 3].

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Received: 11.02.2026 Accepted: 25.02.2026 Published: 30.03.2026

CONCLUSION

Hypotheses — outcomes

H1 — PRIMARY

Dual-path multi-task net achieves $\text{MAE} < 2.5^\circ$ on leakage-mitigated splits.

CONFIRMED

Val MAE $2.19^\circ \pm 0.01$ (3 seeds) · holdout 2.482° + TTA.
Best run 2.17° at 512 px with ConvNeXt-Tiny.

H2 — SECONDARY

Learned uncertainty ranks high-error cases for selective expert review.

CONFIRMED

Top-10% lowest- σ subset $\rightarrow$ **1.77°** MAE. Calibrated intervals require post-hoc scaling and prospective validation.

H3 — EXPLORATORY

Adaptable to clinical domain via limited fine-tuning; zero-shot is constrained.

CONFIRMED

AASCE19 zero-shot $4.85^\circ \rightarrow$ fine-tuned **2.19°** (50 epochs, lr $5e-5$). 92.17% within 5° of expert.

Thank you
for your attention.

I welcome your questions and discussion.

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